

PRARABDHA ALE MAGAR THAPA

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EDUCATION

Rutgers University

Bachelor of Arts in Computer Science

New Brunswick, NJ

Sep. 2021 - May 2025

Coursework

Artificial Intelligence, Software Methodology, Algorithms, Principles of Information & Data Management, Data Science

EXPERIENCE

Software Engineer Intern (Full Stack Developer)

New York, NY

Bunchful Enterprise

March 2025 - May 2025

- Built an SEO audit website using **Google Search Console API**, analyzing backlinks, accessibility, and search metrics to optimize live websites, improving Bunchful's SEO scores by **15%**.
- Developed a full-stack eCommerce platform using **Next.js** with **Stripe API** integration, enabling a secure payment process and simulating real-time checkout for prospective clients.
- Designed a data collection onboarding portal using **TypeScript**, implementing form validation and storing user inputs into a **MySQL** database, logging **100+** entries during testing.
- Contributed to frontend development, performance testing, and bug resolutions across **5+** Bunchful production projects, streamlining development workflows, and improving deployment readiness.

Website Consultant & Integration Operations

Margate, NJ

Scotts Dock via Celerant

June 2025 - Present

- Collaborated with Celerant to design a new eCommerce website, integrating Big Rock's inventory API to sync **60,000+ SKUs**, while advising on navigation structure, back-office configuration, and site architecture, resulting in **40+ online eCommerce transactions** and **\$1,300+** in revenue since late-summer launch.
- Integrated marine rental software into the homepage, improving user experience and enabling **35+** weekly reservations directly through the website during summer season.
- Analyzed in-store sales data for **2,000+** items using the Cumulus back-office, generating data-driven insights that guided inventory placement, product rotations, and online merchandising layout.

PROJECTS

A* Pathfinding | Java

February 2025

- Built an AI-based pathfinding system in Java initiating Repeated Forward, Repeated Backward, and Adaptive A* algorithms to evaluate efficiency and scalability across **50+** tested grids, applying techniques relevant to robotics, games development and GPS navigation.
- Utilized heuristics, priority queues, and tie-breaking strategies to optimize routes in randomly generated **101x101** mazes, reducing search time for an average of **0.0026** seconds to completion and demonstrating the impact of compared algorithmic design to refine performance.

PathToPeace | Python, Flask, SQLite, scikit-learn LR

March 2025

- Generated a machine learning model to assist patients in identifying PTSD risk by utilizing scikit-learn, pandas, and NumPy to analyze **1,400+** PTSD survey entries and achieve a **92%** predictive accuracy when diagnosing new at-risk patients through age, sex, and PCL-5 test results.
- Integrated the linear regression model into a Flask-based web application with **SQLite-backed data storage** to securely log prediction results with a personalized graph to visualize progress and features such as anonymous patient messaging and clinician dashboards for administrative oversight and feedback.

Image Recognition | Python, NumPy, pickle

May 2025

- Established and trained a customizable **three-layer neural network** for image recognition and binary classification, enabling users to train their own datasets with adjustable input size and sample sets for flexible AI applications, tested on **1,000+** sample facial and digit datasets with an **89%** average accuracy.
- Implemented **forward pass**, **backpropagation**, and **gradient descent** with Xavier initialization to enhance weight distribution and prevent vanishing gradients, reducing model training time from several minutes to an average **16.5 seconds** across sample datasets, and exported models with pickle for easy reuse for future applications.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: Spring Boot, React Native, Next.js, Flask, Node.js, Express.js

Libraries & Tools: pandas, NumPy, scikit-learn, Git, VS Code, IntelliJ, Firebase, SQLite, MySQL, PostgreSQL